Unit 12: Derivatives of Trig Functions — Full Solutions

Warm-up

1. $y = \sin x + \cos x$

$$y' = \cos x + (-\sin x) = \cos x - \sin x$$

2. $y = 3\sin x - 2\cos x$

$$y' = 3\cos x - 2(-\sin x) = 3\cos x + 2\sin x$$

3. $y = \tan x + 5$

$$y' = \sec^2 x + 0 = \sec^2 x$$

4. $y = 4\cos x - x^2$

$$y' = 4(-\sin x) - 2x = -4\sin x - 2x$$

5. $y = \sec x + \cot x$

$$y' = \sec x \tan x + (-\csc^2 x) = \sec x \tan x - \csc^2 x$$

6. $y = 2\csc x - 3\tan x$

$$y' = 2(-\csc x \cot x) - 3\sec^2 x = -2\csc x \cot x - 3\sec^2 x$$

7. $y = x + \sin x$

$$y' = 1 + \cos x$$

Standard

8. $y = x\sin x$

Let $f = x$ ($f' = 1$), $g = \sin x$ ($g' = \cos x$).

$$y' = 1 \cdot \sin x + x \cdot \cos x = \sin x + x\cos x$$

9. $y = x^2 \cos x$

Let $f = x^2$ ($f' = 2x$), $g = \cos x$ ($g' = -\sin x$).

$$y' = 2x\cos x + x^2(-\sin x) = 2x\cos x - x^2\sin x$$

10. $y = \dfrac{\sin x}{x}$

Let $f = \sin x$ ($f' = \cos x$), $g = x$ ($g' = 1$).

$$y' = \frac{\cos x \cdot x - \sin x \cdot 1}{x^2} = \frac{x\cos x - \sin x}{x^2}$$

11. $y = (x+1)\cos x$

Let $f = x+1$ ($f' = 1$), $g = \cos x$ ($g' = -\sin x$).

$$y' = 1 \cdot \cos x + (x+1)(-\sin x) = \cos x - (x+1)\sin x$$

12. $y = \dfrac{\tan x}{x}$

Let $f = \tan x$ ($f' = \sec^2 x$), $g = x$ ($g' = 1$).

$$y' = \frac{\sec^2 x \cdot x - \tan x \cdot 1}{x^2} = \frac{x\sec^2 x - \tan x}{x^2}$$

13. $y = \sin x\cos x$

Let $f = \sin x$ ($f' = \cos x$), $g = \cos x$ ($g' = -\sin x$).

$$y' = \cos x \cdot \cos x + \sin x \cdot (-\sin x) = \cos^2 x - \sin^2 x$$

14. $y = x\cos x - \sin x$

Differentiate the product $x\cos x$ using the product rule: $f = x$ ($f' = 1$), $g = \cos x$ ($g' = -\sin x$).

$$\frac{d}{dx}[x\cos x] = 1 \cdot \cos x + x(-\sin x) = \cos x - x\sin x$$

Differentiate the remaining term: $\dfrac{d}{dx}[-\sin x] = -\cos x$.

Add them together:

$$y' = (\cos x - x\sin x) + (-\cos x) = -x\sin x$$

$y' = -x\sin x$. (Notice the $\cos x$ terms cancelled out completely!)

Challenge

15. $y = (\sin x + \cos x)(x^2 + 2x)$

Let $f = \sin x + \cos x$ ($f' = \cos x - \sin x$), $g = x^2 + 2x$ ($g' = 2x + 2$).

$$y' = (\cos x - \sin x)(x^2 + 2x) + (\sin x + \cos x)(2x + 2)$$

This is already a complete, correct answer. If you'd like to expand it fully, here's how it simplifies:

Expand the first piece: $(\cos x - \sin x)(x^2 + 2x) = x^2 \cos x + 2x \cos x - x^2 \sin x - 2x \sin x$.

Expand the second piece: $(\sin x + \cos x)(2x + 2) = 2x \sin x + 2 \sin x + 2x \cos x + 2 \cos x$.

Add everything and combine like terms (the $-2x \sin x$ and $+2x \sin x$ cancel):

$$y' = x^2 \cos x - x^2 \sin x + 4x \cos x + 2 \sin x + 2 \cos x$$

$y' = (\cos x - \sin x)(x^2 + 2x) + (\sin x + \cos x)(2x + 2)$, which simplifies to $x^2 \cos x - x^2 \sin x + 4x \cos x + 2 \cos x + 2 \sin x$.

16. $y = x^2 \sin x + 2x \cos x - 2 \sin x$

Differentiate each of the three terms separately.

Term 1 ($x^2 \sin x$, product rule with $f = x^2$, $f' = 2x$, $g = \sin x$, $g' = \cos x$):

$$\frac{d}{dx}[x^2 \sin x] = 2x \sin x + x^2 \cos x$$

Term 2 ($2x \cos x$, product rule with $f = 2x$, $f' = 2$, $g = \cos x$, $g' = -\sin x$):

$$\frac{d}{dx}[2x \cos x] = 2 \cos x + 2x(-\sin x) = 2 \cos x - 2x \sin x$$

Term 3 ($-2 \sin x$):

$$\frac{d}{dx}[-2 \sin x] = -2 \cos x$$

Add all three results together:

$$y' = (2x \sin x + x^2 \cos x) + (2 \cos x - 2x \sin x) + (-2 \cos x)$$

Combine like terms: the $2x \sin x$ and $-2x \sin x$ cancel; the $2 \cos x$ and $-2 \cos x$ cancel.

$$y' = x^2 \cos x$$

$y' = x^2 \cos x$. Everything else cancelled out beautifully.

17. Deriving $\dfrac{d}{dx}[\tan x]$ from $\tan x = \dfrac{\sin x}{\cos x}$.

Let $f = \sin x$ ($f' = \cos x$) and $g = \cos x$ ($g' = -\sin x$). Apply the quotient rule:

$$\frac{d}{dx}\left[\frac{\sin x}{\cos x}\right] = \frac{\cos x \cdot \cos x - \sin x \cdot (-\sin x)}{(\cos x)^2}$$

Simplify the top:

$$\cos x \cdot \cos x - \sin x \cdot (-\sin x) = \cos^2 x + \sin^2 x$$

By the Pythagorean identity, $\sin^2 x + \cos^2 x = 1$, so the top simplifies to just 1:

$$\frac{1}{\cos^2 x}$$

Since $\dfrac{1}{\cos x} = \sec x$, we have $\dfrac{1}{\cos^2 x} = \sec^2 x$.

Answer: $\dfrac{d}{dx}[\tan x] = \sec^2 x$ — confirming the formula from the toolbox, built entirely from the quotient rule and one identity.

18. $y = \sec x \tan x$

Let $f = \sec x$ ($f' = \sec x \tan x$), $g = \tan x$ ($g' = \sec^2 x$).

$$y' = (\sec x \tan x)(\tan x) + (\sec x)(\sec^2 x) = \sec x \tan^2 x + \sec^3 x$$

$y' = \sec x \tan^2 x + \sec^3 x$. (This can also be rewritten using $\tan^2 x = \sec^2 x - 1$ as $2\sec^3 x - \sec x$, but either form is a correct final answer.)

19. $y = \dfrac{x \sin x}{\cos x}$

Simplify first, using $\dfrac{\sin x}{\cos x} = \tan x$:

$$y = x \cdot \frac{\sin x}{\cos x} = x \tan x$$

Now apply the product rule with $f = x$ ($f' = 1$), $g = \tan x$ ($g' = \sec^2 x$):

$$y' = 1 \cdot \tan x + x \cdot \sec^2 x = \tan x + x \sec^2 x$$

$y' = \tan x + x \sec^2 x$. (Simplifying first turned a messy quotient-rule problem into a quick product-rule one.)

20. $y = \csc x \cot x$

Let $f = \csc x$ ($f' = -\csc x \cot x$), $g = \cot x$ ($g' = -\csc^2 x$).

$$y' = (-\csc x \cot x)(\cot x) + (\csc x)(-\csc^2 x) = -\csc x \cot^2 x - \csc^3 x$$

$y' = -\csc x \cot^2 x - \csc^3 x$.

Applied

21. $x(t) = 5\sin t$

$$v(t) = x'(t) = 5\cos t$$

$v(0) = 5\cos(0) = 5(1) = 5$.

$v\left(\frac{\pi}{2}\right) = 5\cos\left(\frac{\pi}{2}\right) = 5(0) = 0$.

In words: at $t = 0$, the pendulum is passing through the center of its swing (where $x(0) = 0$) at its maximum speed of 5 cm/s. At $t = \frac{\pi}{2}$, the pendulum is momentarily at rest ($v = 0$) — this is the instant it reaches the far edge of its swing (where $x\left(\frac{\pi}{2}\right) = 5\sin\left(\frac{\pi}{2}\right) = 5$, its maximum displacement), right before it swings back.

22. $I(t) = 100 + 20\cos t$

$$I'(t) = -20\sin t$$

$$I'\left(\frac{\pi}{2}\right) = -20\sin\left(\frac{\pi}{2}\right) = -20(1) = -20$$

Answer: $I'(t) = -20\sin t$, and $I'\left(\frac{\pi}{2}\right) = -20$. This means that at $t = \frac{\pi}{2}$ hours, sunlight intensity is decreasing at a rate of 20 units per hour.

23. $h(t) = 10 + 8\sin t$

$$h'(t) = 8\cos t$$

$$h'(0) = 8\cos(0) = 8(1) = 8$$

Answer: $h'(t) = 8\cos t$, and $h'(0) = 8$. Since $h'(0) = 8$ is positive, the rider is rising at $t = 0$, moving upward at a rate of 8 meters per unit of time.

24. $V(t) = 5\cos t + 2\sin t$

$$V'(t) = -5\sin t + 2\cos t$$

$$V'(0) = -5\sin(0) + 2\cos(0) = 0 + 2(1) = 2$$

Answer: $V'(t) = -5\sin t + 2\cos t$, and $V'(0) = 2$. This means that at $t = 0$, the voltage is increasing at a rate of 2 volts per unit of time.